

**Question 1.1.** At least \_\_\_\_\_% of the people are between 25 and 65 years old. Explain your answer!  
**(6 Points)**

93.75%



**Question 1.2.** At most \_\_\_\_\_% of the people have ages that are not in the range 25 years to 65 years.  
Explain your answer! **(6 Points)**

6.25%



**Question 1.3.** At most \_\_\_\_\_% of the people are more than 65 years old. Explain your answer! (6 Points)

*Hint:* If you're stuck, try thinking about what the distribution may look like in this case.

6.25%



**Question 2.2.** Suppose the data science class decides to construct a 90% confidence interval instead of a 95% confidence interval, but they still require that the width of the interval is no more than 6% from left end to right end. Will they need the same sample size as in 2.1? Pick the right answer and explain further without calculation. **(6 Points)**

1. Yes, they must use the same sample size.
2. No, a smaller sample size will work.
3. No, they will need a bigger sample.



**Question 2.4.** This shows that the percentage in a normal distribution that is at most 1.65 SDs above average is about **95%**. Explain why 1.65 is the right number of SDs to use when constructing a **90%** confidence interval. **(6 Points)**

because 5% in each tail  $\Rightarrow$  1.65 SDs will capture 90% of the normal distribution  $\Rightarrow$  1.65 is the right number of SDs to use when constructing a 90% confidence interval



**Question 3.2.** Why does the Central Limit Theorem (CLT) apply in this situation, and how does it explain the distribution we see above? We recommend reviewing [14.4](#) for a refresher on CLT. **(6 points)**

we can see from the distribution, it's follow normal distribution

Because 400 is a large number, we also draw sample with replacement for 10000 times => the distribution follows normal distribution

